Medical Microbiology

Third stage- College of Dentistry

Lab. 10

Clostridium

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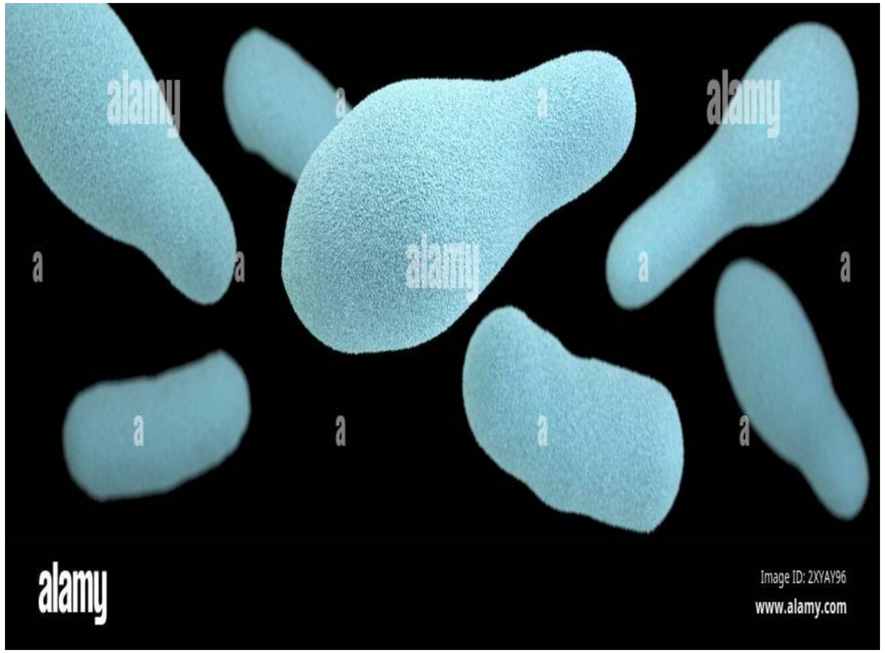
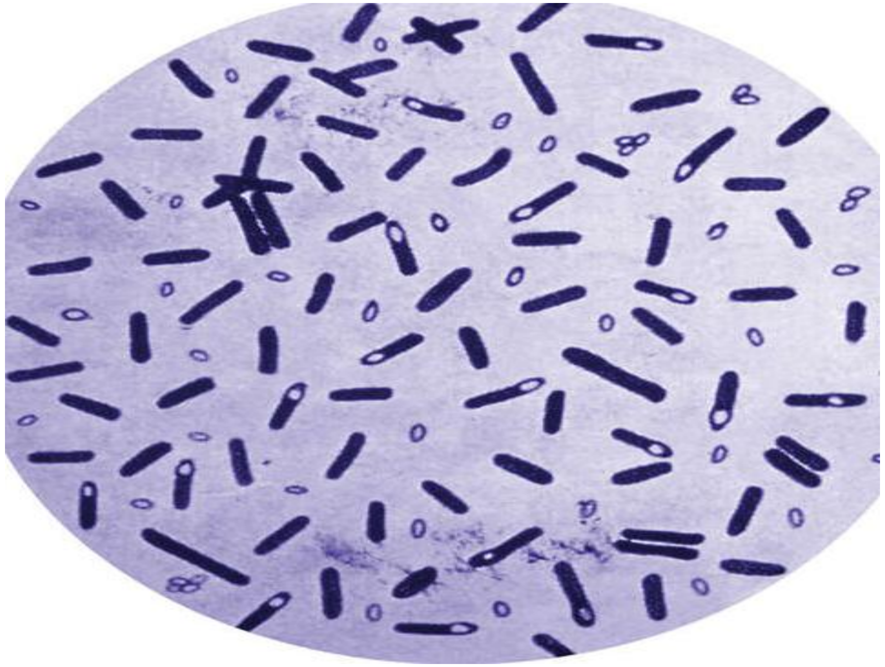
Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic appearance of *Clostridium*.
2-Explain the laboratory methods used to detect Clostridium spp.
3- Explain types of biochemical test for detect of Clostridium.
4- Identify the different diseases associated with each species of *Clostridium*.





General characteristics of *Clostridium*

- **Shape:** Rod-shaped (bacilli)
- **Gram Reaction:** Gram-positive
- **Oxygen Requirement:** Obligate anaerobes
- **Motility:** Mostly motile, **except *C. perfringens***
- **Capsule:** Generally non-capsulated, **except *C. perfringens***
- **Metabolism:** Fermentative in nature.

Clostridium ferments:

-Carbohydrates → organic acids + alcohols + gases.

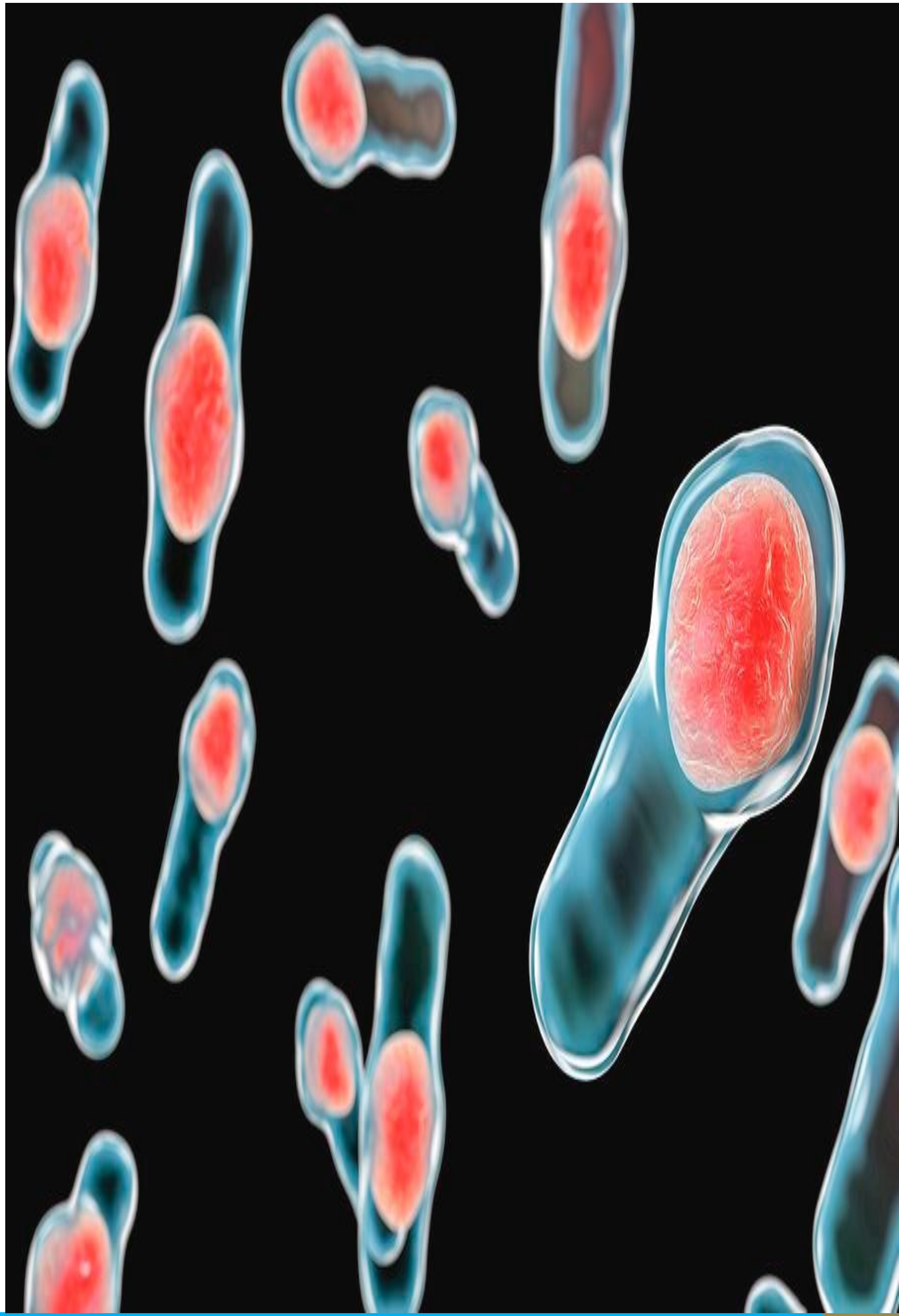
-Amino acids (in some species).

This fermentation process provides energy under **anaerobic conditions**.

Spore Formation:

Clostridium species form tough spore, metabolically inactive endospores that are highly resistant to heat, drying, and disinfectants.

The spores are typically wider than the bacterial rod and may be located centrally, sub terminally, or terminally within the cell.



Natural Habitat:

Clostridium species commonly inhabit soil, water, and the gastrointestinal tracts of humans and animals, where their spores are widely distributed.

Uses:

C. butyricum: A non-pathogenic species present in the gut, soil, and certain dairy products. It functions as a probiotic and produces butyrate (butyric acid), which supports intestinal health.

C. thermocellum: Studied for its ability to break down **lignocellulosic biomass** and convert it into **ethanol**, making it a promising organism for use in biofuel generation.

Other Species

- ***Clostridium botulinum***

Appears as **pleomorphic Gram-positive rods** that can vary in shape. Forms **oval, sub terminal spores** Produces botulinum toxin (Botox), which is used clinically to treat muscle spasms and other medical conditions. The spores are highly resistant and allow the organism to survive in soil and canned foods. **Motile**, possessing peritrichous flagella.

- ***Clostridium tetani***

Characterized by **long, slender Gram-positive rods**.
Produces a **terminal, spherical spore** that gives the classic **"drumstick" or "tennis racket" appearance** due to marked swelling at one end of the cell.
Motile with peritrichous flagella.
Typically found in soil and the intestinal tracts of animals.

- ***Clostridium perfringens***

Appears as **large, pleomorphic Gram-positive rods** with blunt ends. Forms **oval, sub terminal spores**, but spores are not commonly seen in clinical specimens.

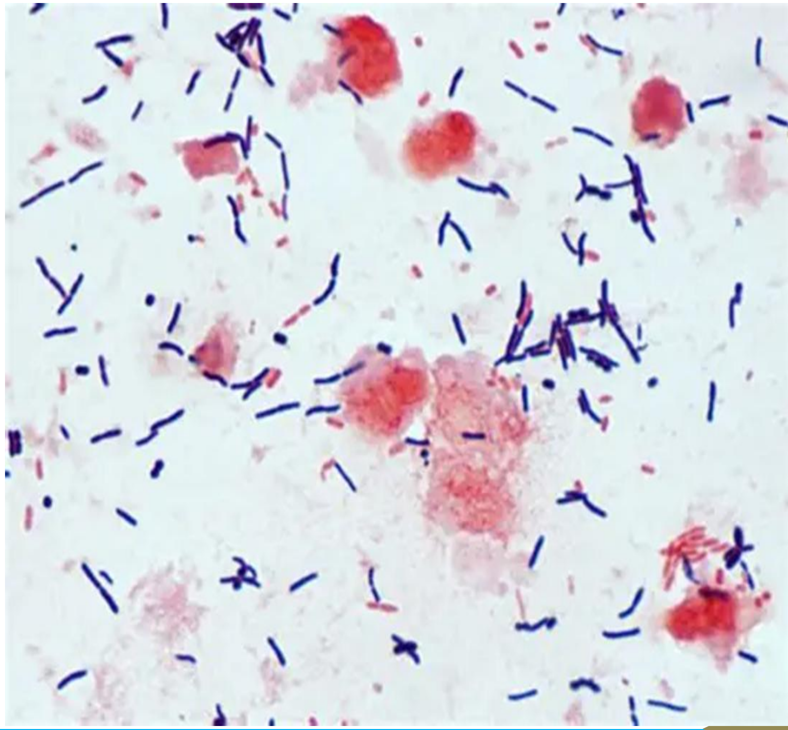
Non-motile, which helps distinguish it from many other Clostridium species. Grows rapidly and is known for producing **double-zone hemolysis** on blood agar due to its potent toxins.

- ***Clostridioides (Clostridium) difficile***

Displays **long, thin Gram-positive rods**. Produces **large, oval, sub terminal spores** that often cause a noticeable bulge in the cell.

Motile, possessing peritrichous flagella.

Spores are highly resistant and allow the organism to persist in the environment and healthcare settings. Known for causing **antibiotic-associated colitis** due to overgrowth when normal flora are disrupted.



Culture Characteristics of Clostridium Species:

- ***Clostridium botulinum***

Blood agar: Forms **large, semi-transparent colonies** with an irregular or **wavy margin**. Most strains exhibit **β-hemolysis**, producing a clear zone of hemolysis around the colonies. Colonies may take **24–48 hours** to become visible under anaerobic conditions.

- ***Clostridium tetani***

Blood agar: Grows as a **fine, spreading film** across the agar surface rather than forming discrete colonies.

On **fresh blood agar plates**, it initially shows **α-hemolysis** (partial hemolysis), which is usually followed by **β-hemolysis** (complete hemolysis) after prolonged incubation.

Requires **strict anaerobic conditions** for optimal growth, and colonies are typically slow-growing.

- ***Clostridium perfringens***

Blood Agar Plates (BAP): Cultured **anaerobically**, producing characteristic colonies, often showing **double-zone hemolysis** due to alpha and beta toxins.

Tryptose Sulfite Cycloserine (TSC) Agar: A **selective and differential medium** that promotes growth of *C. perfringens* while inhibiting other bacteria; colonies often appear **black due to sulfite reduction**.

Thioglycolate Broth: A **reducing medium** used to culture anaerobes and assess growth patterns under low oxygen conditions.

This combination of media allows for **isolation and differentiation** of *C. perfringens* from clinical or environmental samples.

- ***Clostridioides (Clostridium) difficile:***

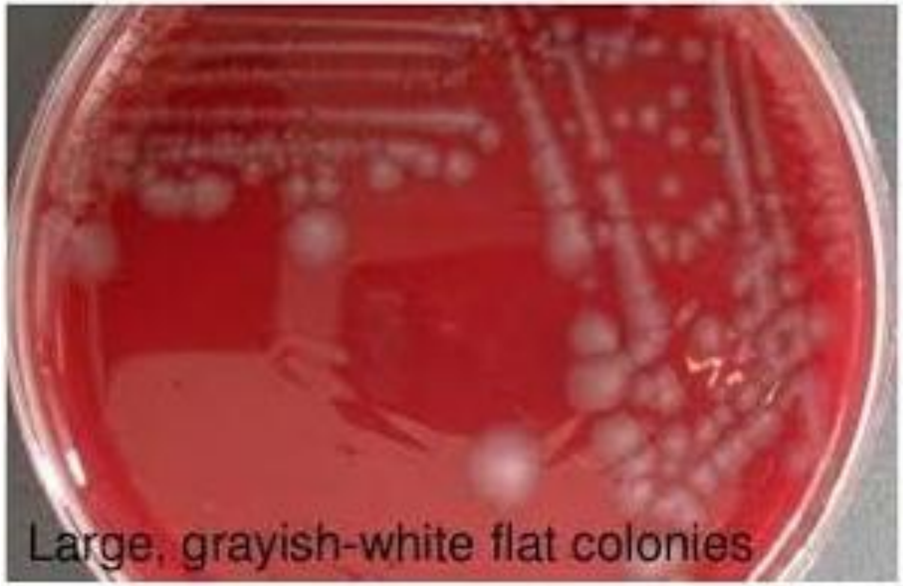
Blood Agar Plates (BAP, anaerobic conditions): Forms **large, flat colonies** with a distinctive **barnyard or horse-stable odor**.

Cycloserine Cefoxitin Fructose Agar (CCFA, selective medium):

Used to **selectively isolate C. difficile** from mixed flora.

Colonies are typically **~4 mm in diameter**, **yellow**, with a **ground-glass appearance** and **filamentous edges**.

These media allow for both **isolation and identification** based on colony morphology and odor.



Blood agar

CCFA

Biochemical Tests of some species of Clostridium:

- **Indole Test:**

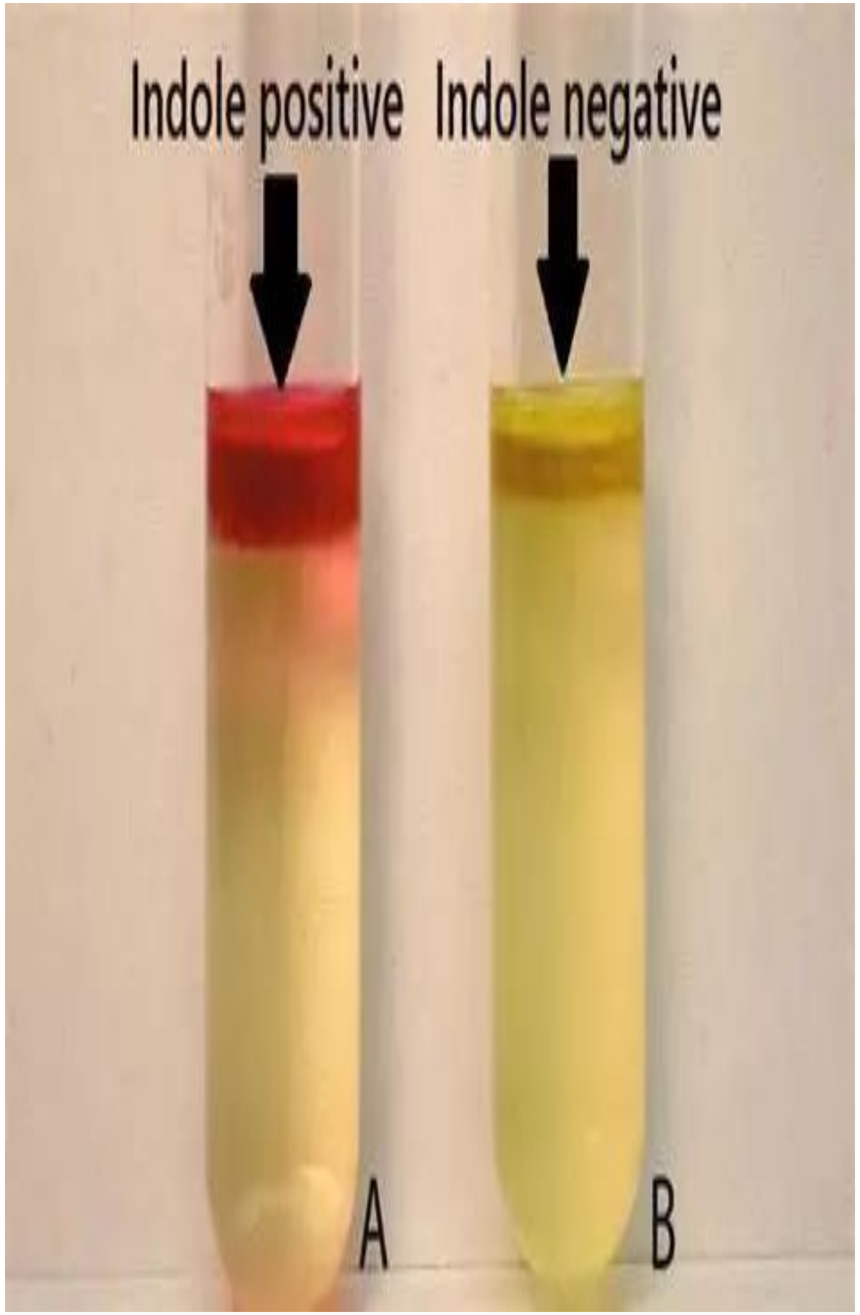
Purpose: Detects the ability of *Clostridium tetani* to produce **indole** from **tryptophan**.

Procedure: *C. tetani* is inoculated into **tryptophan broth** and incubated anaerobically. After incubation, **Kovac's reagent** is added to the culture.

Interpretation:

Positive result: Formation of a **red-colored ring** at the surface of the broth indicates **indole production**.

Negative result: No color change, indicating the organism cannot produce indole.



• Bile Esculin Test:

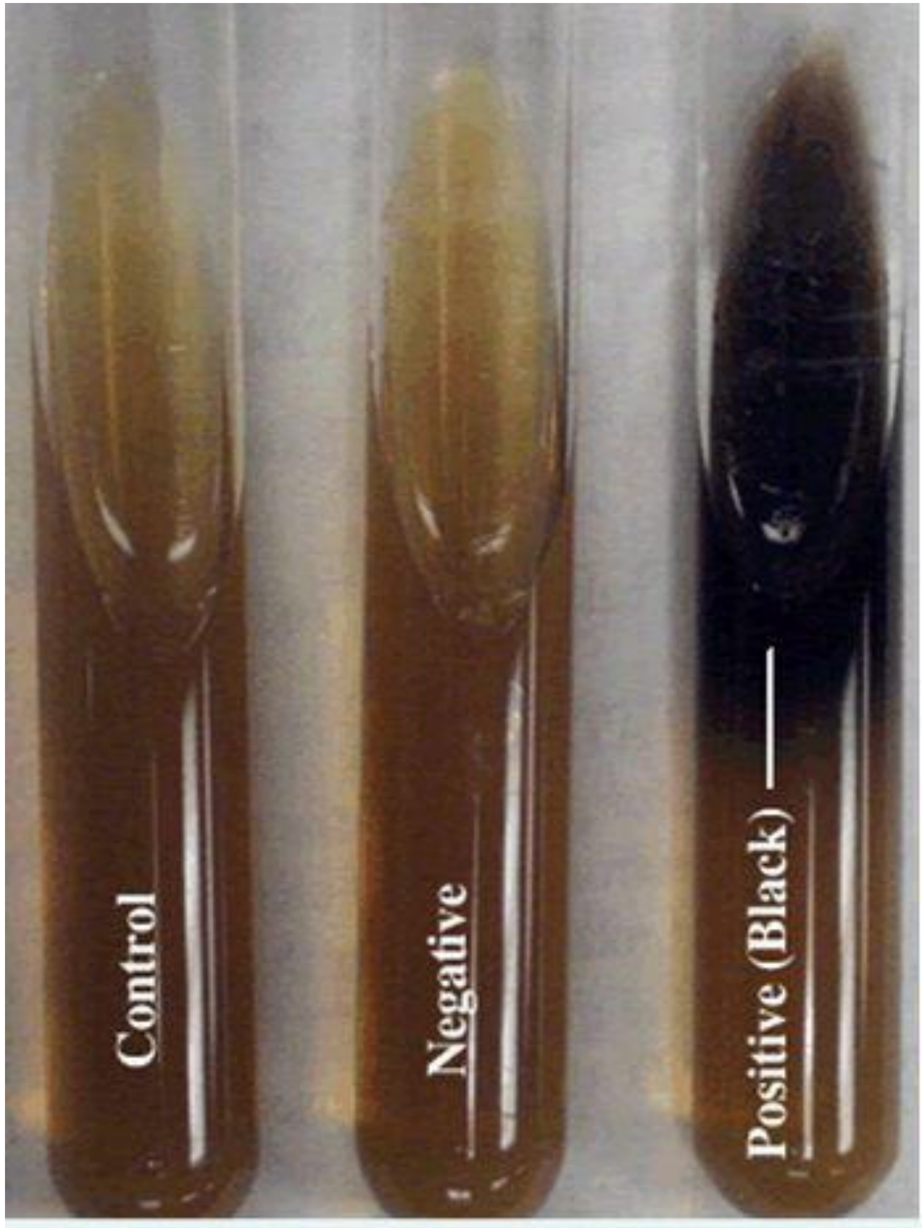
Purpose: Detects the ability of *Clostridioides (Clostridium) difficile* to **hydrolyze esculin** in the presence of bile.

Procedure: *C. difficile* is inoculated onto **bile esculin agar** and incubated anaerobically.

Interpretation:

Positive result: Hydrolysis of esculin produces **a black coloration** in the medium, indicating the organism can grow in the presence of bile and hydrolyze esculin.

Negative result: No color change, indicating inability to hydrolyze esculin.



• Gelatin Liquefaction Test:

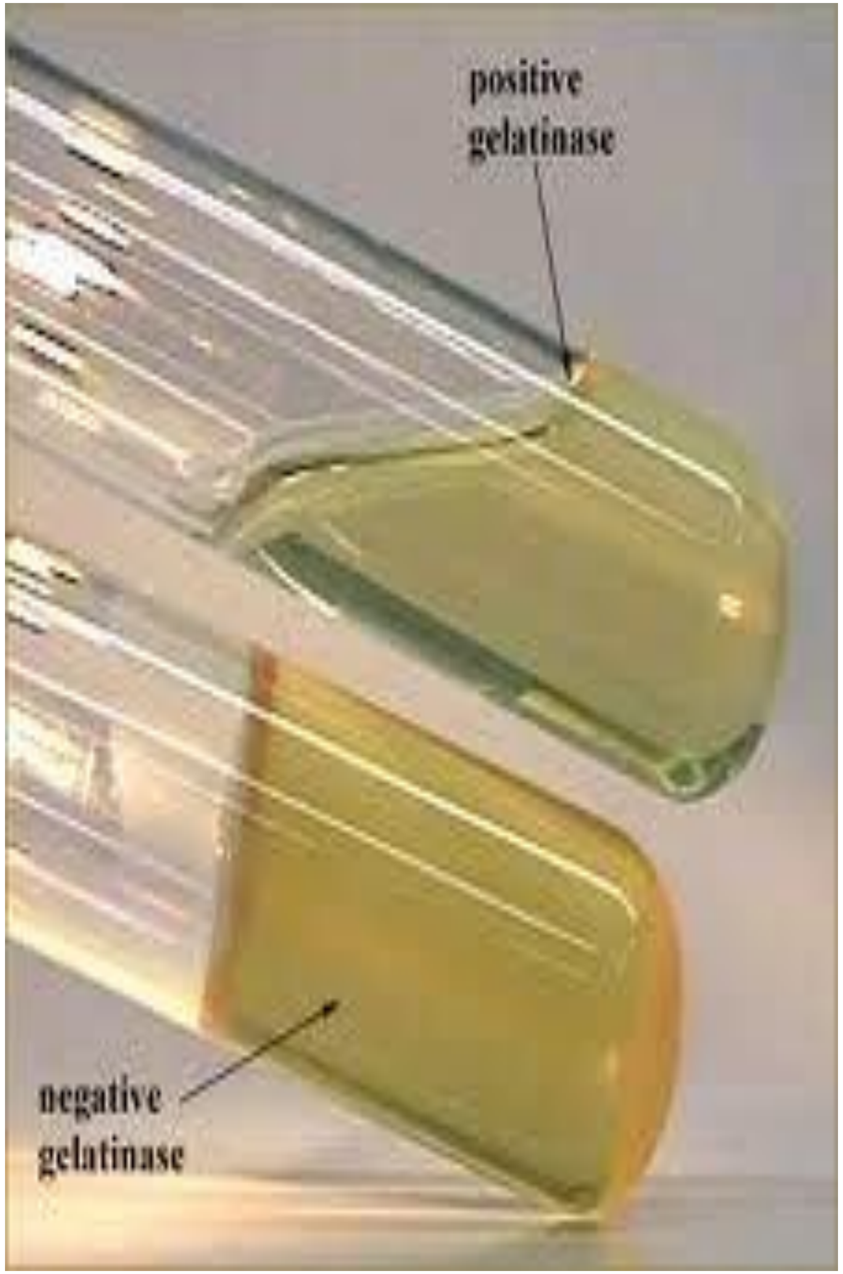
Purpose: Detects the ability of *Clostridium perfringens* to produce **gelatinase**, a proteolytic enzyme that hydrolyzes gelatin.

Procedure: *C. perfringens* is inoculated into **gelatin-containing medium** and incubated under anaerobic conditions.

Interpretation:

Positive result: The gelatin **liquefies**, indicating production of gelatinase.

Negative result: The gelatin **remains solid**, indicating the organism does not produce gelatinase.



Pathogenesis

Species	Primary Disease	Notes on Toxin/Infection
C. botulinum	Botulism	Produces the Botulinum Toxin (Botox), causing flaccid paralysis.
C. tetani	Tetanus	Produces Tetanospasmin, which causes spastic paralysis or muscle rigidity ("lockjaw").
C. perfringens	Gas Gangrene and Food Poisoning	Causes severe tissue necrosis. Gas gangrene occurs when it enters deep wounds and proliferates in the absence of oxygen
C. difficile / Clostridioides difficile	Pseudomembranous Colitis and Diarrhea	Thrives when antibiotics kill beneficial gut bacteria, releasing toxins that damage the colon lining

Tetanus

Botulism

Pseudomembranous Colitis and Diarrhea

Gas Gangrene

Any Question?

Thank you